

Problem Set 4
Due: Mar. 22, 2021

1. Section 2.2 Problem 4.

An internet service provider uses 50 modems to serve the needs of 1000 customers. It is estimated that at a given time, each customer will need a connection with probability 0.01, independent of the other customers.

- (a). What is the PMF of the number of modems in use at the given time?
- (b). Repeat part (a) by approximating the PMF of the number of customers that need a connection with a Poisson PMF.
- (c). What is the probability that there are more customers needing a connection than there are modems? Provide an exact, as well as an approximate formula based on the Poisson approximation of part (b).

Solution:

- (a). Let X be the number of modems in use. For $k < 50$, the probability that $X = k$ is the same as the probability that k out of 1000 customers need a connection:

$$p_X(k) = \binom{1000}{k} 0.01^k 0.99^{1000-k}, \quad k = 0, 1, \dots, 49$$

The probability that $X = 50$, is the same as the probability that 50 or more out of 1000 customers need a connection:

$$p_X(50) = \sum_{k=50}^{1000} \binom{1000}{k} 0.01^k 0.99^{1000-k}$$

- (b). By approximating the binomial in (a) with a Poisson with parameter $\lambda = 1000 \cdot 0.01 = 10$, we have

$$p_X(k) = e^{-10} \frac{10^k}{k!}, \quad k = 0, 1, \dots, 49,$$

$$p_X(50) = \sum_{k=50}^{1000} e^{-10} \frac{10^k}{k!}.$$

Or, if you notice that this problem has changed the random variable to the number of “customers”, you will get

$$p_X(k) = e^{-10} \frac{10^k}{k!}, \quad k = 0, 1, \dots, 1000$$

- (c). Let A be the event that there are more customers needing a connection than there are modems. Then,

$$\mathbf{P}(A) = \sum_{k=51}^{1000} \binom{1000}{k} 0.01^k 0.99^{1000-k}$$

With the Poisson approximation, $\mathbf{P}(A)$ is estimated by

$$\mathbf{P}(A) = \sum_{k=51}^{1000} e^{-10} \frac{10^k}{k!}$$

2. Section 2.3 Problem 16.

Let X be a random variable with PMF

$$p_X(x) = \begin{cases} x^2/a & \text{if } x = -3, -2, -1, 0, 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a). Find a and $\mathbf{E}[X]$.
(b). What is the PMF of the random variable $Z = (X - \mathbf{E}[X])^2$?
(c). Using the result from part (b), find the variance of X .
(d). Find the variance of X using the formula $\text{var}(X) = \sum_x (x - \mathbf{E}[X])^2 p_X(x)$.

Solution:

- (a). The scalar a must satisfy the normalization axiom,

$$1 = \sum_x p_X(x) = \frac{1}{a} \sum_{x=-3}^3 x^2$$

So,

$$a = \sum_{x=-3}^3 x^2 = 28$$

We also have $\mathbf{E}[X] = 0$ because the PMF is symmetric around 0.

- (b). If $z \in \{1, 4, 9\}$, then

$$p_Z(z) = p_X(\sqrt{z}) + p_X(-\sqrt{z}) = \frac{z}{28} + \frac{z}{28} = \frac{z}{14}$$

Otherwise, $p_Z(z) = 0$.

- (c).

$$\text{var}(X) = \mathbf{E}[Z] = \sum_z z p_Z(z) = \sum_{z \in \{1, 4, 9\}} \frac{z^2}{14} = 7$$

(d).

$$\begin{aligned}\text{var}(X) &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= 1^2 \cdot (p_X(-1) + p_X(1)) + 2^2 \cdot (p_X(-2) + p_X(2)) + 3^2 \cdot (p_X(-3) + p_X(3)) \\ &= 2 \cdot \frac{1}{28} + 8 \cdot \frac{4}{28} + 18 \cdot \frac{9}{28} \\ &= 7\end{aligned}$$

3. Suppose that two teams play a series of games that ends when one of them has won 3 games. Suppose that each game played is, independently, won by team A with probability p . Find the expected number of games that are played. Also, show that this number is maximized when $p = 1/2$.

Solution: Let N denote the number of games played. Suppose A wins. If only 3 games are played, the result must be three successive wins for A. If 4 games are played, A wins 2 of the first 3 games and the fourth game. If 5 games are played, A wins 2 of the first 4 games and the fifth game. Suppose B wins, the analysis is similar. Thus,

$$\begin{aligned}\mathbf{E}[N] &= 3 \cdot [p^3 + (1-p)^3] + 4 \cdot [3p^2(1-p)p + 3p(1-p)^2(1-p)] + 5 \cdot [6p^2(1-p)^2p + 6p^2(1-p)^2(1-p)] \\ &= 6p^4 - 12p^3 + 3p^2 + 3p + 3\end{aligned}$$

Differentiation yields

$$\frac{d}{dp} \mathbf{E}[N] = 24p^3 - 36p^2 + 6p + 3$$

which is 0 at $p = 1/2$.

4. Let X be a RV such that

$$\begin{cases} p_X(1) = p \\ p_X(-1) = 1 - p \end{cases}$$

Find $c \neq 1$, such that $\mathbf{E}[c^X] = 1$.

Solution:

$$\mathbf{E}[c^X] = \sum_x c^x p_X(x) = cp + c^{-1}(1-p) = 1$$

Hence,

$$pc^2 - c + 1 - p = 0$$

Equivalently,

$$(pc - 1 + p)(c - 1) = 0$$

Thus,

$$c = \frac{1-p}{p}$$